

Meridian Valley General Hospital

Facilities Engineering & Plant Operations Facilities Engineering Operations & Maintenance Manual

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1. Purpose and Scope

This manual establishes operating procedures for the Facilities Engineering & Plant Operations department, responsible for the physical plant, utility systems, life-safety systems, and maintenance of all buildings at Meridian Valley General Hospital (Main Hospital, Medical Office Building, Central Utility Plant, and Parking Structure).

2. Work Order System

All maintenance requests are submitted through the Computerized Maintenance Management System (CMMS) at cmms.meridianvalley-demo.org or by calling the Facilities Help Desk (ext. 2200). Requests are triaged by priority:

Priority	Response Time	Examples
Emergency (P1)	Immediate / <15 min	Loss of power, medical gas failure, active flooding, fire
Urgent (P2)	<2 hours	HVAC failure in patient area, elevator entrapment, no hot water
Routine (P3)	<48 hours	Broken chair, flickering light, minor leak, door adjustment
Preventive/Scheduled	Per PM calendar	Filter changes, generator testing, inspections

3. Critical Utility Systems

3.1 Emergency Power

Three 1,500 kW diesel generators provide backup power to life-safety branch, critical branch, and equipment branch circuits per NFPA 99/110. Generators are load-tested monthly (30 min under load) and undergo a full 4-hour annual load bank test. Automatic transfer switches (ATS) are inspected quarterly. Fuel reserves are maintained for a minimum 96-hour run time.

3.2 Medical Gas Systems

Oxygen, medical air, and vacuum systems are monitored 24/7 via the Building Automation System (BAS) with alarm panels at the Plant Operations control room and Nursing Supervisor station. Master alarm signals must be investigated within 10 minutes. Annual certification testing is performed per NFPA 99 by a certified medical gas verifier.

3.3 HVAC and Air Handling

Operating rooms and isolation rooms are monitored for pressure relationships (positive for OR/protective isolation, negative for airborne infection isolation) with continuous differential pressure monitors. Air changes per hour (ACH) are validated semi-annually: OR minimum 20 ACH (4 fresh), All rooms minimum 12 ACH.

4. Preventive Maintenance (PM) Program

Equipment/System	Frequency	Responsible Team
Emergency Generators	Monthly load test / Annual load bank	Plant Operations - Electrical
Fire Alarm & Sprinkler Systems	Quarterly / Annual (NFPA 25/72)	Life Safety Team + Contractor
Medical Gas Alarm Panels	Monthly function test	Plant Operations - Mechanical
Air Handling Units (AHUs)	Quarterly filter change, annual overhaul	HVAC Shop
Elevators (9 total)	Monthly inspection, annual state cert.	Contracted Elevator Vendor
Nurse Call Systems	Semi-annual test	Biomedical/IT Liaison
Boilers & Chillers	Weekly checks, annual inspection	Central Plant Team
Eyewash Stations	Weekly flush test	Facilities Safety Officer

5. Life Safety and Interim Life Safety Measures (ILSM)

Any construction, renovation, or system impairment that affects fire/life safety triggers an ILSM assessment (Form ENG-LS-04) before work begins, covering temporary egress routes, fire watch requirements, dust/infection control barriers (per ICRA matrix), and notification of Nursing Administration and the Safety Officer.

6. Infection Control Risk Assessment (ICRA) for Construction

All construction and renovation projects require an ICRA permit classifying the project (Type A-D) and required precautions (Class I-IV), jointly signed off by Facilities, Infection Prevention, and the affected unit manager before work begins.

7. Utility Failure Response

- Power failure: Confirm generator start and ATS transfer within 10 seconds; notify House Supervisor and Nursing Administration immediately; log event in CMMS.
- Water disruption: Activate Water Outage Plan (bottled water carts, portable handwash stations); notify Infection Prevention and affected units.
- Medical gas alarm: Dispatch engineer to source location within 10 minutes; do not silence alarm until cause confirmed and corrected.
- HVAC failure in OR/isolation room: Suspend use of affected room until pressure relationships and air changes are restored and verified.

8. Contact

Facilities Engineering & Plant Operations, Sub-Basement Level 1 | Help Desk: ext. 2200 (24/7) | Emergency
Pager: 555-0142 | Email: facilities@meridianvalley-demo.org